Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	**AI-Powered Smart Agriculture Monitoring and Crop Disease Prediction Platform**	
S. No.	**University ID**	**Name**
1	2420090026	B Vaishnavi
2	2420090055	S Lahari Krishna
Name of the Guide		Ms.G Lavanya

Abstract

AI-Powered Smart Agriculture Monitoring and Crop Disease Prediction Platform

AgriShield AI is an AI-powered smart agriculture monitoring and crop disease prediction platform developed to help farmers identify crop health issues at an early stage and make informed decisions. The system allows users to upload images of crop leaves, which are processed by a machine learning model to detect possible diseases and provide a confidence score. Based on the detected condition, the platform displays basic preventive measures and recommendations. A centralized dashboard provides useful insights such as total crop scans, healthy and diseased crops, disease distribution, prediction history, and overall farm health.

The project integrates **DevOps and MLOps practices** to ensure efficient development, testing, deployment, and maintenance. **FastAPI** is used to build scalable backend APIs, while **PostgreSQL** stores prediction records and monitoring data. **MLflow** is incorporated for tracking machine learning experiments, evaluation metrics, and model versions. **Docker** is used to containerize the application and provide a consistent execution environment. **GitHub Actions** automates testing, Docker image building, and continuous integration and deployment. **Jira/Trello** supports Agile project planning, task tracking, and team collaboration.

Additionally, the platform maintains a **prediction timeline** that helps users observe changes in crop health over time. The system can also display model and service status, providing better visibility into the reliability of the application. By combining **machine learning, web technologies, database management, MLOps, and DevOps automation**, AgriShield AI provides a practical, scalable, and user-friendly solution for intelligent crop monitoring and early disease detection.

Signature of the Guide HOD